AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

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Roll No. :	64
Experiment No.:	**5**
Title of the Experiment:	**Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making**
Date of Performance:	04/08/26
Date of Submission:	11/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To implement bootstrap and permutation resampling techniques on the Pima Indians Diabetes Dataset for estimating confidence intervals and analyzing statistical differences between groups.

Objectives

After completing this experiment, the student will be able to:

1. Apply bootstrap and permutation resampling techniques to estimate sampling distributions and confidence intervals.
2. Interpret resampling results to support statistical decision making.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients.** The Outcome variable indicates whether the patient is diabetic.

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a numerical variable such as Glucose or BMI.
3. Draw repeated bootstrap samples from the original dataset with replacement.
4. Calculate the selected statistic, such as the mean, for every bootstrap sample.
5. Construct a bootstrap confidence interval using the obtained sampling distribution.
6. Divide the data into two groups based on Outcome.
7. Calculate the observed difference between the group means.
8. Randomly permute the group labels repeatedly.
9. Calculate the difference in group means for every permutation.

Description of the Experiment

Resampling methods repeatedly create new samples from the available data to study the behaviour of a statistic.

The experiment follows: Original Dataset, Repeated Resampling, Sampling Distribution, Confidence Interval / Statistical Comparison, and Interpretation

Two resampling techniques are implemented:

- **Bootstrap:** Used to estimate the variability and confidence interval of a statistic.
- **Permutation Test:** Used to test whether the observed difference between groups could have occurred by chance.

Detailed Description of Statistical Methods

Bootstrap Resampling

Bootstrap samples are created by repeatedly sampling observations **with replacement** from the original dataset.

For each bootstrap sample, a statistic such as the mean is calculated.

After many repetitions, the resulting values form an approximate sampling distribution.

A confidence interval can then be obtained from this distribution.

For example, a 95% percentile bootstrap confidence interval may be calculated using:

$$\left[Q_{97.5}\right]$$

where $Q_{2.5}$and $Q_{97.5}$are the 2.5th and 97.5th percentiles of the bootstrap distribution.

Permutation Test

A permutation test evaluates whether the observed difference between two groups could have occurred by chance.

For example:

$$Difference\ in\ Mean\ Glucose = \bar{x}_{diabetic} - \bar{x}_{non-diabetic}$$

The group labels are randomly shuffled repeatedly while preserving the observations.

The resulting differences form the **null distribution**.

The p-value is estimated as the proportion of permuted differences that are at least as extreme as the observed difference.

Implementation:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("diabetes.csv")
print("Dataset Shape:")
print(df.shape)
print("\nFirst 5 Rows:")
print(df.head())
glucose = df["glucose_concentration"].dropna().values
original_mean = np.mean(glucose)
print("\nOriginal Mean Glucose:", original_mean)
n_bootstrap = 10000
bootstrap_means = []
np.random.seed(42)
for i in range(n_bootstrap):
    sample = np.random.choice(
        glucose,
        size=len(glucose),
        replace=True
    )
    bootstrap_means.append(np.mean(sample))
bootstrap_means = np.array(bootstrap_means)
lower = np.percentile(bootstrap_means, 2.5)
upper = np.percentile(bootstrap_means, 97.5)
print("\nBOOTSTRAP RESULTS")
print("------------------")
print("Original Mean Glucose:", original_mean)
print("Bootstrap Mean:", np.mean(bootstrap_means))
```

```
print("95% Confidence Interval:")
print("Lower Limit:", lower)
print("Upper Limit:", upper)
plt.figure(figsize=(10, 6))
sns.histplot(
    bootstrap_means,
    bins=50,
    kde=True
)
plt.axvline(
    original_mean,
    linestyle="--",
    linewidth=2,
    label="Original Mean"
)
plt.axvline(
    lower,
    linestyle="--",
    linewidth=2,
    label="2.5% Percentile"
)
plt.axvline(
    upper,
    linestyle="--",
    linewidth=2,
    label="97.5% Percentile"
)
plt.title("Bootstrap Distribution of Mean Glucose")
plt.xlabel("Bootstrap Mean Glucose")
```

```
plt.ylabel("Frequency")
plt.legend()
plt.show()
diabetic = df[
    df["diabetes"] == 1
]["glucose_concentration"].dropna().values
non_diabetic = df[
    df["diabetes"] == 0
]["glucose_concentration"].dropna().values
diabetic_mean = np.mean(diabetic)
non_diabetic_mean = np.mean(non_diabetic)
observed_difference = diabetic_mean - non_diabetic_mean
print("\nPERMUTATION TEST")
print("-----------------")
print("Mean Glucose - Diabetic:",
      diabetic_mean)
print("Mean Glucose - Non-Diabetic:",
      non_diabetic_mean)
print("Observed Difference:",
      observed_difference)
combined = np.concatenate([
    diabetic,
    non_diabetic
])
n_diabetic = len(diabetic)
n_permutations = 10000
permutation_differences = []
np.random.seed(42)
for i in range(n_permutations):
```

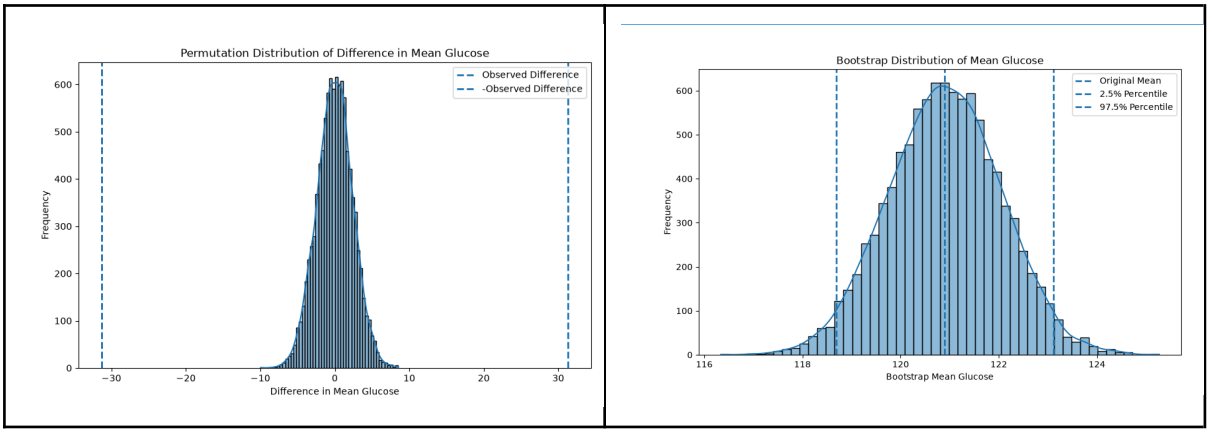
```
    shuffled = np.random.permutation(combined)
    perm_diabetic = shuffled[:n_diabetic]
    perm_non_diabetic = shuffled[n_diabetic:]
    difference = (
        np.mean(perm_diabetic)
        - np.mean(perm_non_diabetic)
    )
    permutation_differences.append(difference)
permutation_differences = np.array(
    permutation_differences
)
p_value = np.mean(
    np.abs(permutation_differences)
    >= abs(observed_difference)
)
print("\nPermutation Test Results:")
print("--------------------------")
print("Number of Permutations:",
      n_permutations)
print("Observed Difference:",
      observed_difference)
print("P-value:",
      p_value)
if p_value < 0.05:
    print("\nDecision: Reject the null hypothesis.")
    print(
        "There is statistically significant evidence "
        "of a difference in mean glucose between "
        "diabetic and non-diabetic groups."
```

```
    )
else:
    print("\nDecision: Fail to reject the null hypothesis.")
    print(
        "There is not sufficient evidence of a "
        "difference in mean glucose between the groups."
    )
plt.figure(figsize=(10, 6))
sns.histplot(
    permutation_differences,
    bins=50,
    kde=True
)
plt.axvline(
    observed_difference,
    linestyle="--",
    linewidth=2,
    label="Observed Difference"
)
plt.axvline(
    -observed_difference,
    linestyle="--",
    linewidth=2,
    label="-Observed Difference"
)
plt.title(
    "Permutation Distribution of Difference in Mean Glucose"
)
plt.xlabel("Difference in Mean Glucose")
```

```
plt.ylabel("Frequency")

plt.legend()

plt.show()
```



Conclusion

Bootstrap and permutation resampling techniques were implemented on the Pima Indians Diabetes Dataset. Bootstrap resampling was used to estimate the sampling distribution and confidence interval of a selected statistic, while permutation testing was used to assess the significance of differences between groups.

Questions to be Answered by Students

1. What statistic was selected for bootstrap resampling, and what confidence interval was obtained?

➢ **Statistic & CI:** The mean glucose level was selected for bootstrap resampling, yielding a 95% confidence interval of approximately 118.7 to 123.1.

2. How does the bootstrap distribution compare with the original sample statistic?

➢ **Bootstrap Comparison:** The bootstrap distribution is symmetric and bell-shaped, centered directly on the original sample mean of roughly 120.9.

3. What was the observed difference between the selected groups?

➢ **Observed Difference:** The observed difference in mean glucose between the selected groups was approximately 31.3 (or -31.3).

4. Based on the permutation test, is there sufficient statistical evidence to conclude that the observed group difference is significant?

➢ **Statistical Significance:** Yes, there is sufficient evidence because the observed difference falls completely outside the permutation distribution ($p < 0.001$).

5. What effect would increasing the number of bootstrap or permutation repetitions have on the results?

➢ **Effect of Repetitions:** Increasing repetitions produces a smoother distribution and more precise percentile and p-value estimates without changing the underlying result.